

Example 1

The time in years for a personal computer to function before breaking down is a continuous random variable with probability density function

$$f(x) = \begin{cases} \lambda e^{-x/9} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

in which λ is a constant.

- a) Determine λ .
- b) What is the expected lifetime of the computer?
- c) What is the probability that it will function less than 6 years? Discuss whether it agrees with your expectation when buying a PC.
- d) What is the name of this distribution? This distribution has a famous property. What is it? Do you think this property is realistic?

Note: Integration by parts is needed for part b). Below is a revision of integration by parts:

Integration by parts stems from the product rule for differentiation. If u and v are functions of x , then

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Integration with respect to x gives

$$\int \frac{d}{dx}(uv) dx = \int u \frac{dv}{dx} dx + v \frac{du}{dx} dx$$

Therefore

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

Hence

$$\int u dv = uv - \int v du$$

Answers

a) The area under the curve is 1. Hence

$$\int_0^{\infty} \lambda e^{-\frac{x}{9}} dx = 1 \Rightarrow 9\lambda = 1 \Rightarrow \lambda = \frac{1}{9}$$

b) The expected value is

$$E[X] = \int_0^{\infty} x \left(\frac{1}{9}\right) e^{-x/9} dx$$

Using integration by part,

$$u = -x \quad dv = e^{-\frac{x}{9}} d\left(-\frac{x}{9}\right) \Rightarrow v = e^{-x/9}$$

$$E[X] = -xe^{-\frac{x}{9}} \Big|_0^{\infty} - \int_0^{\infty} e^{-\frac{x}{9}} d(-x) = 9 \text{ years}$$

Note: Revision of integration by parts:

Integration by parts stems from the product rule for differentiation. If u and v are functions of x , then

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Integration with respect to x gives

$$\int \frac{d}{dx}(uv) dx = \int u \frac{dv}{dx} dx + v \frac{du}{dx} dx$$

Therefore

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

Hence

$$\int u dv = uv - \int v du$$

c)

$$\int_0^6 \left(\frac{1}{9}\right) e^{-\frac{x}{9}} dx = -\left(e^{-\frac{6}{9}} - 1\right) = 0.486582881 < 0.5$$

There are more than 50% chance that the computer will survive at least 6 years.

d)

The distribution in the question is known as the exponential distribution.

The exponential distribution has the famous property of being memoryless.

A non-negative random variable X is memoryless if

$$P\{X > s + t | X > t\} = P\{X > s\} \quad \text{for all } s, t \geq 0$$

It is not fully realistic because the computer components will age. So the computer should be more and more prone to break down given the same period of time s as the computer is getting old.

Also, if the quality control is not good, the chance of a new computer breaking down may also be higher.

Example 2

The number of attempts before achieving the first success can be considered as a discrete random variable with a probability mass function. Consider the following question.

The probability of a violin student getting the Distinction in an examination is 0.4. If the student wishes to get a Distinction, what is the expected number of attempts that the student takes the examination? Discuss the assumption(s) made behind your calculations.

Answers

Geometric Distribution.

Since $p = 0.4$

The expected number of examinations is $1/p = 2.5 \rightarrow 3$

It makes the assumption that each attempt is independent. In reality, the attempts are not independent. For example, if the first attempt does not get Distinction, the student has meanwhile gained experience in calmly taking violin examination. Then the second attempt will have a higher probability of getting Distinction, or the student may attempt a violin examination at a higher grade, etc.

Example 3

In Hong Kong, criminal cases are tried usually with a 7 person jury. In a particular case, suppose that the jurors are independent and based on the evidence, each is inclined to give a verdict of “innocent” with a probability of 0.75 in a vote, what is the probability that the vote will be a majority verdict of “innocent”?

A majority verdict is 7 – 0, 6 – 1 or 5 – 2.

Answers

$$\binom{7}{7} (0.75)^7 + \binom{7}{6} (0.75)^6 (0.25) + \binom{7}{5} (0.75)^5 (0.25)^2 = 0.756408691$$

Example 4

The following web site tests your typing speed: <https://www.typingtest.com/test.html>

Set up a probability model for the occurrence of mistyped words. What are your underlying assumptions?

Answers

A word is either typed correctly or mistyped. Assume that the number of typed words is large and the probability of mistyped words is low. The assumptions fit the requirement of Poisson distribution.

Repeat the typing tests a few times. Set λ = average no. of mistyped words.

Example 5

Suppose the average number of accidents occurring weekly on a particular stretch of a highway equals 3. Calculate the probability that there is at least one accident this week.

Answers

Assume it is a Poisson distribution.

$$P\{X \geq 1\} = 1 - P\{X = 0\} = 1 - e^{-3} \frac{3^0}{0!} = 1 - e^{-3} \approx 0.9502$$

Example 6

Buses arrive at a specified stop at 15-minute intervals starting at 7 a.m. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits

- a) less than 5 minutes for a bus;
- b) at least 12 minutes for a bus

Answers

- a) Let X denote the time in minutes past 7 a.m. that the passenger arrives at the stop. Since X is a uniform random variable over the interval $(0, 30)$, the passenger has to wait less than 5 minutes if he arrives between 7:10 and 7:15 or between 7:25 and 7:30.

$$P\{10 < X < 15\} + P\{25 < X < 30\} = \frac{5}{30} + \frac{5}{30} = \frac{1}{3}$$

- b) Similarly, he has to wait for at least 12 minutes if he arrives between 7:00 and 7:03 or between 7:15 and 7:18.

$$P\{0 < X < 3\} + P\{15 < X < 18\} = \frac{3}{30} + \frac{3}{30} = \frac{1}{5}$$

Example 7

A tennis player has a probability of winning a tournament of 0.1. Assume that the probability of winning remain unchanged for each tournament.

- a) What is the probability that he wins 5 tournament titles after playing 20 tournaments?
- b) What is the probability that he wins the 5th tournament title at the 20th tournament he enters?
- c) How many matches is he expected to play before winning 5 tournaments?

Answers

- a) This situation can be modelled by the binomial random variable.

$$P\{X = 5\} = \binom{20}{5} (0.1)^5 (0.9)^{15} = 3.1921361\%$$

- b) This situation can be modelled by the negative binomial random variable.

$$P\{X = 5\} = \binom{19}{4} (0.1)^5 (0.9)^{15} = 0.798034028 \%$$

- c) Let X_i be the random variable denoting the number of tournaments played before winning the i^{th} tournament after $(i - 1)$ tournaments have been won.

$$E[X_1 + \cdots + X_r] = E[X_1] + \cdots + E[X_r]$$

Since

$$E[X_i] = \frac{1}{p}$$

$$E[X] = \frac{r}{p} = \frac{5}{0.1} = 50$$

The player is expected to play 50 tournaments.

Example 8

A buyer of components buys them in lots of size 100. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are non-defective. Suppose the manufacturing process has some systematic error such that in every lot, there is 1 defective component, what proportion of lots does the buyer reject?

Answers

The hypergeometric random variable is used.

Let E be the event that the buyer accepts the lot.

$$P(E) = \frac{\binom{99}{3} \binom{1}{0}}{\binom{100}{3}} = 0.97$$

So he would reject 3% of the lots he inspected.